QP CODE: 25901712

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC)–MG2DSCHPY100-PHYSIOLOGY

OF HUMAN CONTROL SYSTEM

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Interest (I), Appreciation(Ap), and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple ChoiceQuestions
Answer all questions
Each question carries **1** mark

1. Option A – Ciliary body
2. Option C - Axoplasm
3. Option A -Thyroid releasing Hormone
4. Option A - Neurotransmitter
5. Option A - Hormones
6. Option D - Hypoparathyroidism

Part B

Short Answer Type Questions.
Answer any **2** questions.
Each question carries **2** marks.

7.Define the term Mastication [K][CO1]

Mastication is the process of chewing,where food is broken down into smaller pieces (

It involves both mechanical and chemical processes.**(1 mark)**

8. **Discuss the sensory function of nervous system [U] [CO2]**

- The nervous system detects changes in the environment (both internal and external) through sensory receptors. (**1 mark)**
- Sensory neurons transmit signals to the brain and spinal cord. **(1 mark))**

9. **Discuss the term Diabetes Mellitus [U]/[CO2]**

Deficiency of insulin results in an elevation of blood glucose level.Increased levels of glucose in the blood results in its elimination in urine along with large amounts of water.The excessive loss of water through urine creates unusual thirst and the consequent increased intake of water.These abnormalities in glucose metabolism and associated symptoms due to the deficiency of insulin are collectively called diabetes mellitus.**(2 marks)**

10. **Discuss the hormones secreted by Adrenal cortex.[U]/[CO3]**

Mineralocorticoids are secreted by the outermost region of the adrenal cortex.The principal mineralocorticoid is aldosterone,which controls the salt and water balance in the body.**(1/2 mark)**

Glucocorticoids are secreted by the middle region of the adrenal cortex.The principal glucocorticoid is cortisol,which increases blood glucose levels.The other glucocorticoids are cortisone and corticosterone.**(1 mark)**

Gonadocorticoids are secreted by the inner region of the adrenal cortex.These include the male and female sex hormones androgen and oestrogen.**(1/2 mark)**

Part C

Short Essay Type Questions.
Answer any **4** questions.
Each question carries **5** marks.

11. Explain auditory defects [U] [CO1]

Auditory Fatigue (1/4 mark)

- It is the transient loss of sensitivity due to prolonged hearing of a loud sound. **(1/4 mark)**

- The fatigue is binaural (relating to both ear) because if one ear is subjected to prolonged sound the other is not, then both the ears suffer the loss of sensitivity. **(1/4 mark)**
- If the frequency of sound is 1000 cycles per second or above, then it produces fatigue more intensely than that of frequency below. **(1/4 mark)**

Noise

- It is described in psychological terms as a sound which causes disturbance or annoyance. **(1/4 mark)**
- If the auditory apparatus is exposed to continued noise, the mechanism of hearing is seriously affected and this may lead to loss of hearing. **(1/4 mark)**
- The degree of loss of hearing depends upon the frequency value of the noise. **(1/4 mark)**
- After continued exposure to industrial noises the sensitivity of hearing is found to be diminished or lost. **(1/4 mark)**

Masking(1/4 mark)

- Weaker sounds may be completely inaudible in presence of louder sounds. This phenomenon is called masking. **(1/4 mark)**
- The sounds of almost same frequency cause fluctuations in loudness. **(1/4 mark)**
- The whole length of the basilar membrane is affected by low tones, where as a limited region is affected by high tones and that is why the low tones are more effective in masking. **(1/4 mark)**

Deafness

- The primary acoustic centre is in the temporal lobe of the cerebrum. Removal of both temporal lobes is followed by complete deafness and of one temporal lobe is followed by impairment of hearing. Deafness may be of two types: **(1/4 mark)**

1. **Conductive deafness**

In conductive deafness there is interference with the passage of sound waves through the external ear and middle ear. **(1/4 mark)**

a) **External ear obstruction:** The conductive deafness occurs due to entrance of foreign bodies, or due to hard or dry wax in the external ear. The damage or perforation of tympanic membrane may be the cause of failure of conduction. **(1/4 mark)**

b) **Middle ear disease:** Any condition which prevents the normal functioning of the ossicles. **(1/4 mark)**

2. Nerve Deafness

- The nerve deafness is due to loss of function of organ of Corti and also due to interference of transmission of impulses by the auditory nerve. The temporary nerve deafness occurs after exposure to a very loud sound. The main causes are: **(1/4 mark)**

  i) Due to bacterial or viral infection as in meningitis in children.

  ii) Due to acoustic trauma as in boiler-makers.

  iii) Due to toxic action of the drugs, viz. streptomycin, quinine, measles, etc.

  iv) Due to pressure of a tumour at the junction of cerebellum and pons. **(any 3 points 1/4 mark each)**

12. Explain reflex activity [U] [CO2]

- Reflex activity is the response to a peripheral nervous stimulation that occurs without our consciousness. (**1/2 mark)**
- It is a type of protective mechanism and it protects the body from irreparable damages. (**1/2 mark)**

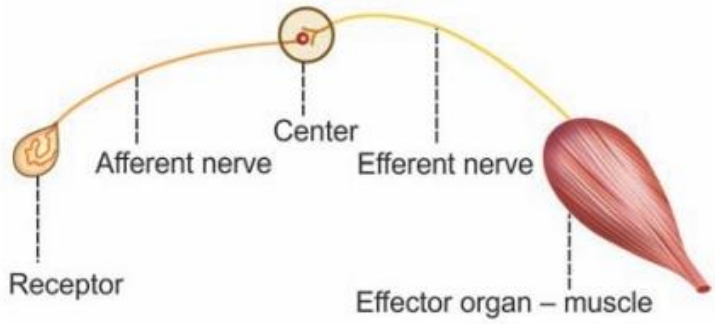
Reflex Arc

Reflex arc is the anatomical nervous pathway for a reflex action. A simple reflex arc includes five components. (**1/2 mark)**

1. **Receptor :** Receptor is the end organ, which receives the stimulus. When receptor is stimulated, impulses are generated in afferent nerve. (**1/2 mark)**
2. **Afferent Nerve**: Afferent or sensory nerve transmits sensory impulses from the receptor to centre. (**1/2 mark)**

3. **Centre**: Centre receives the sensory impulses via afferent nerve fibers and in turn it generates appropriate motor impulses. Centre is located in the brain or spinal cord. (**1/2 mark)**
4. **Efferent Nerve**: Efferent or motor nerve transmits motor impulses from the centre to the effector organ. (**1/2 mark)**
5. **Effector Organ**: Effector organ is the structure such as muscle or gland where the activity occurs in response to stimulus. (**1/2 mark)**

Diagram (**1 mark)**



13. Explain the role of Testes[U]/[CO3]

1) **Sperm Production(Spermatogenesis)**-This occurs within the seminiferous tubules of the testes,where spermatogonia divide and differentiate into mature sperm**(1 mark)**
2) **Production of Testosterone**-Testosterone is the primary male sex hormone,produced by Leydig cells located in the interstitial spaces between the seminiferous tubules.**(1 mark)**
3) **Regulation of Male Reproductive System**-Testosterone and other hormones are involved in the regulation of hypothalamic-pituitary-gonadal axis,which controls sperm production and hormone secretion.**(1 mark)**
4) **Storage and Maturation of Sperm**-Sperm produced in the seminiferous tubules are transported to the epididymis,a coiled tube located on the back of the testes,where they mature and are stored until ejaculation.**(1 mark)**

5)Regulation of Sexual Function-Testosterone influences libido(sexual desire)and is crucial for maintaining erectile function.It also contributes to the production of semen.**(1 mark)**

14. Describe the parts of the Olfactory System[U][CO1]

1)Nasal Epithelium - Also called Olfactory Epithelium.Specialized epithelial tissue inside the nasal cavity that is involved with the sense of smell.Part of the olfactory system directly responsible for detecting odours **(1 mark)**

2)Receptor cells- Olfactory receptor cells in the cell membranes of olfactory receptor neurons are responsible for the detection of odour molecules.Activated olfactory receptors are the initial player in a signal cascade,which ultimately produces a nerve impulse that is transmitted to the brain **(1 mark)**

3)Glomerulus - Spherical structure in the olfactory bulb.The glomerulus is the basic unit in the odour map of the olfactory bulb.Each odour activates a different pattern of glomeruli,which decode the identity of the odour.**(1 mark)**

4)Mitral Cells - The mitral cells are neurons that are part of the Olfactory system.Located in the olfactory bulb in the mammalian central nervous system.Receive information from the axons of olfactory receptor neurons,forming synapses in neuropils called glomeruli.**(1 mark)**

5)Olfactory bulb- Present on the inferior side of the brain.Divided into two distinct structures,the main olfactory bulb and the accessory olfactory bulb.The Olfactory bulb transmits smell information from the nose to the brain.**(1 mark)**

15. Classify nerve fibres

1. **Depending Upon Structure**
   i. **Myelinated Nerve Fibers**: The nerve fibers that are covered by myelin sheath.
   ii. **Non-myelinated Nerve Fibers**: The nerve fibers which are not covered by myelin sheath
2. **Depending Upon Distribution**
   i. **Somatic Nerve Fibers**: Somatic nerve fibers supply the skeletal muscles of the body.

ii. **Visceral or Autonomic Nerve Fibers**: Autonomic nerve fibers supply the various internal organs of the body.

3. **Depending Upon Origin**
   i. **Cranial Nerve Fibers** : Nerve fibers arising from brain
   ii. **Spinal Nerve Fibers** : Nerve fibers arising from spinal cord
4. **Depending Upon Function**
   i. **Sensory Nerve Fibers**: Sensory nerve fibers carry sensory impulses from different parts of the body to the central nervous system. These nerve fibers are also known as afferent nerve fibers.
   ii. **Motor Nerve Fibers**: Motor nerve fibers carry motor impulses from central nervous system to different parts of the body. These nerve fibers are also called efferent nerve fibers.
5. **Depending Upon Secretion of Neurotransmitter**
   i. **Adrenergic Nerve Fibers**: Adrenergic nerve fibers secrete noradrenaline.
   ii. **Cholinergic Nerve Fibers**: Cholinergic nerve fibers secrete acetylcholine.

(1/2 mark for each point)

16.Explain the hormones secreted by the Thyroid gland[U][CO3]

Thyroid Gland produces two hormones-**thyroxine(T4) and triiodothyronine(T3),**which are iodinated derivatives of the amino acid ,thyrosine.The active hormone is thyroxine.Triiodothyronine is about four times as potent as thyroxine,but it is present in the blood in much smaller quantities and persists for a much shorter time than does thyroxine.**(2 marks)**

Functions of Thyroid hormones: Stimulation of the rate of cellular metabolism resulting in the generation of more heat energy,Differentiation of tissues,growth and skeletal maturation,Thyroxine promotes excessive elimination of water through urine,enhances basal metabolic rate,causes increased glucose utilization and mobilization of glycogen reserves.**(2 marks).**

Calcitonin is another hormone secreted by the parafollicular cells of the Thyroid gland.This hormone opposes the action of the parathyroid glands by reducing the

calcium level in the blood.If blood calcium level becomes too high,calcitonin is secreted until calcium ion level decrease to normal**.(1 mark)**

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the functions of skin[U]/[CO1]

1. **Protective Function**
   - **a)** Protection from Bacteria and Toxic Substances
   - **b)** Protection from Mechanical Blow
   - **c)** Protection from Ultraviolet Rays **(2 marks)**
2. **Sensory Function** - Skin has many nerve endings, which form the specialized cutaneous receptors. These receptors are stimulated by sensations of touch, pain, pressure or temperature sensation and convey these sensations to the brain via afferent nerves.
3. **Storage Function** - Skin stores fat, water, chloride and sugar. It can also store blood by the dilatation of the cutaneous blood vessels.
4. **Synthetic Function** - Vitamin D3 is synthesized in skin by the action of ultraviolet rays from sunlight on cholesterol. Exposure to the sun ensures supply of vitamin D.
5. **Regulation of Body Temperature** - Excess heat is lost from the body through skin by radiation, conduction, convection and evaporation. Cutaneous vasoconstriction diverts the blood to the interior of the body and so diminishes heat loss. Vasodilatation of the skin helps in elimination of heat from the body. Sweat glands of the skin play an active part in heat loss, by secreting sweat. Hairs of skin regulate body temperature by entangling layer of air
6. **Regulation of Water and Electrolyte Balance** - Skin regulates water balance and electrolyte balance by excreting water and salts through sweat
7. **Excretory Function** - Skin excretes small quantities of waste materials like urea, salts and fatty substance through sweat and insensible perspiration.
8. **Absorptive Function** - Lipids are easily permeable through the skin. Lipid-soluble substances like vitamins are easily absorbed through the skin

9. **Secretory Function** - Skin secretes sweat through sweat glands and sebum through sebaceous gland, The sebaceous glands secrete a fatty substance which is rich in cholesterol called sebum. It helps to keep the skin moist, smooth and prevents drying. In cold climates the secretion is depressed and the skin becomes dry and scaly. **(1 mark each for point 2 -9)**

18. Explain the transmission of nerve impulses[U]/[CO2]

Impulse transmission in the nervous system involves the process by which electrical signals or nerve impulses are transferred between neurons or from neurons to muscles or glands. This is a complex process that relies on both electrical and chemical signaling mechanisms. **(1 mark)**

The steps involved are:

1. Resting Membrane Potential
2. Action Potential Initiation
3. Propagation of the Action Potential
4. Synaptic Transmission
5. Postsynaptic Response
6. Reuptake and Degradation of Neurotransmitters **(1 ½ mark for each point)**

19.Explain the functions of Ovaries[U][CO3]

1. Production of Eggs(Oogenesis)- The ovaries produce and release eggs during the menstrual cycle **(2 mark)**

2. **Hormone Production** - **Estrogen**-responsible for the development of female secondary sexual characteristics,promotes the growth of endometrium in preparation for pregnancy-**Progesterone-**prepares the endometrium for the implantation of a fertilized egg and helps maintain early pregnancy,helps regulate the menstrual cycle and prevents further ovulation during pregnancy-**Inhibin -** Helps regulate the levels of FSH,inhibiting excessive egg production and maintaining hormonal balance.**Relaxin-**produced during pregnancy to relax the uterine muscles and the ligaments of the pelvis,facilitating child birth **(1/2 mark for each point)**

3. Regulation of the Menstrual Cycle- The ovaries work in conjuction with hypothalamus and pituitary gland to regulate the menstrual cycle.Hormones like

estrogen and Progesterone are involved in the cyclical changes in the endometrium**(2 mark)**

4. Support of Pregnancy(if fertilization occurs)- The ovaries,continue to produce progesterone to maintain the uterine lining and support the early stages of pregnancy.If pregnancy occurs,the ovaries continue to secrete hormones(including progesterone and estrogen)until the placenta takes over hormone production**(2 mark)**

5.Release of hormones that impact overall health- Estrogen and Progesterone not only regulate the menstrual cycle,but also play a role in bone health,and maintaining the health of the skin,mood and energy levels.The balance of these hormones is critical in preventing conditions such as osteoporosis,heart disease and other metabolic disorders.**(2 mark)**

20. Explain nervous system

Nervous system controls all the activities of the body. It is quicker than other control system in the body. Nervous system is divided into two parts:

CENTRAL NERVOUS SYSTEM

Central nervous system (CNS) includes brain and spinal cord. It is formed by neurons and supporting cells called neuroglia. Structures of brain and spinal cord are arranged in two layers, namely gray matter and white matter. Gray matter is formed by nerve cell bodies and the proximal parts of nerve fibers, arising from nerve cell body. White matter is formed by remaining parts of nerve fibers. **(2 mark)**

In brain, white matter is placed in the inner part and gray matter is placed in the outer part. In spinal cord, white matter is in the outer part and gray matter is in the inner part. Brain is situated in the skull. It is continued as spinal cord in the vertebral canal through the foramen magnum of the skull bone. **(1 mark)**

Brain and spinal cord are surrounded by three layers of meninges called the outer dura mater, middle arachnoid mater and inner pia mater. The space between arachnoid mater and pia mater is known as subarachnoid space. This space is filled with a fluid

called cerebrospinal fluid. Brain and spinal cord are actually suspended in the cerebrospinal fluid. **(1 mark)**

Parts of Brain

1. Prosencephalon

It is otherwise known as forebrain. It is further divided into two parts:

i. Telencephalon, which includes cerebral hemi spheres, basal ganglia, hippocampus and amygda loid nucleus **(1/2 mark)**
ii. Diencephalon, consisting of thalamus, hypothalamus, metathalamus and subthalamus. **(1/2 mark)**

2. Mesencephalon

Mesencephalon is also known as midbrain. **(1/2 mark)**

3. Rhombencephalon

Rhombencephalon or hindbrain is subdivided into two portions:

i. Metencephalon, formed by pons and cerebellum **(1/2 mark)**

ii. Myelencephalon or medulla oblongata. **(1/2 mark)**

Midbrain, pons and medulla oblongata are together called the brainstem. **(1/2 mark)**

PERIPHERAL NERVOUS SYSTEM

Peripheral nervous system (PNS) is formed by neurons and their processes present in all regions of the body. It consists of cranial nerves, arising from brain and spinal nerves, arising from the spinal cord. It is again divided into two subdivisions: (**1 mark)**

1. Somatic Nervous System

Somatic nervous system is concerned with somatic functions. It includes the nerves supplying the skeletal muscles. Somatic nervous system is responsible for muscular activities and movements of the body. **(1 mark)**

2. Autonomic Nervous System

Autonomic nervous system is concerned with regulation of visceral or vegetative functions. So, it is otherwise called vegetative or involuntary nervous system. Autonomic nervous system consists of two divisions, sympathetic division and parasympathetic division. **(1 mark)**